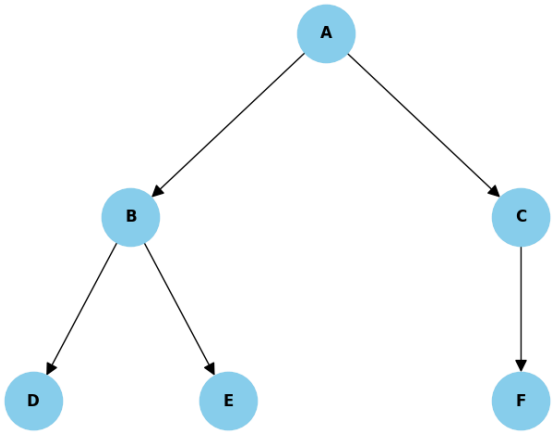
Lab 4

Write a basic program to implement Greedy Best First Search and find the path for the given problem

Example problem



Nodes	Heuristic
A (initial state)	5
B	2
C	3
D	4
E	1
F (goal state)	0

Here, A is the initial state and F is the goal state

Pseudocode

BEGIN

DEFINE adj_list as:
'A' → ['B', 'C']
'B' → ['D', 'E']
'C' → ['F']
'D' → []
'E' → []

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  'F' → []

DEFINE heuristic values:
  'A' → 5
  'B' → 2
  'C' → 3
  'D' → 4
  'E' → 1
  'F' → 0

FUNCTION gbfs(start, goal):

  INITIALIZE visited as empty set
  INITIALIZE priority_queue as empty min-heap
  INSERT (heuristic[start], start) into priority_queue
  INITIALIZE path as empty list

  WHILE priority_queue is not empty:
    POP (heuristic_value, current_node) from priority_queue

    IF current_node IN visited:
      CONTINUE

    APPEND current_node to path
    ADD current_node to visited

    IF current_node == goal:
      BREAK

    FOR each neighbor IN adj_list[current_node]:
      IF neighbor NOT IN visited:
        INSERT (heuristic[neighbor], neighbor) into priority_queue

  RETURN path

FUNCTION display_adjacency_list(adj_list):
  PRINT "Adjacency List:"
  FOR each node, neighbors IN adj_list:
    PRINT node → neighbors

FUNCTION main:
  CALL display_adjacency_list(adj_list)

  INPUT start node
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INPUT destination node

CONVERT both to uppercase

CALL gbfs(start, destination)
STORE result in path

PRINT "Path from [start] to [destination]:"
FOR each node IN path:
PRINT node →
PRINT "End"

CALL main IF this is the main program

END

Report format

Title: Write a program ……..
Date: Lab assigned date

Objective

- To implement the Greedy Best First Search algorithm using an adjacency list to explore nodes in a graph or tree structure.
- To find and display the path from a given start node to a destination node, if such a path exists, using Greedy Best First Search traversal.

Theory

- Greedy Best First Search Algorithm

Implementation

- Write down the problem statement
- Python Code
- Output (Screenshots of your output)

Conclusion